Properties of logarithms

1 Simplify each of the following expressions:

a $\log 4+\log 9$

b $\log_{10}(10)+\log_{10}(10)$

c $\log_{10}11+\log_{10}2+\log_{10}9$

d $\log_{10}12-(\log_{10}2+\log_{10}3)$

e $\dfrac{\log_{10}4}{\log_{10}2}$

f $\dfrac{\log_4 125}{\log_4 5}$

g $\dfrac{\log a^8}{\log a^4}$

h $\dfrac{\log a^3}{\log\sqrt[3]{a}}$

i $\log 10x+\log 10y$

j $\log x^2+\log x^3$

k $\dfrac{\log\left(\dfrac{1}{x^4}\right)}{\log x}$

l $\log_{10}10+\dfrac{\log_{10}\left(15^{20}\right)}{\log_{10}\left(15^{5}\right)}$

m $\dfrac{5\log m^2}{6\log\sqrt[3]{m}}$

n $\dfrac{8\log_{10}\left(\sqrt{10}\right)}{\log_{10}(100)}$

o $10^{\log w}$

p $x^{4\log_x 3-6\log_x 2}$

2 Simplify each of the following expressions:

a $\dfrac{\ln a^6}{\ln a^3}$

b $\dfrac{12\ln\left(m^4\right)}{10\ln\sqrt{m}}$

c $\dfrac{\ln\left(a^4\right)}{\ln\sqrt[3]{a}}$

d $\dfrac{2\ln x}{10\ln y}$

3 Rewrite the following as the sum or difference of logarithms without any powers, fractions or radicals:

a $\log_9 uv$

b $\log\left(x^4\right)$

c $\log_4\left(x^7\right)$

d $\log_5\left(\dfrac{9}{7}\right)$

e $\log\left(x^{\frac{2}{5}}\right)$

f $\log_b\left(x^2\right)$

g $\log\left(3x^{-1}\right)$

h $\log\left((3x)^5\right)$

i $\log\left(7x^{-4}\right)$

j $\log\left((5x)^{-7}\right)$

k $\log\left((2x)^{-1}\right)$

l $\log\left(\dfrac{1}{xy}\right)$

m $\log\left(\dfrac{pq}{r}\right)$

n $\log\left((3x+7)^{-1}\right)$

o $\log\left((3x+4)^{-8}\right)$

p $\log\left((5x+7)^{\frac{1}{2}}\right)$

q $\log\left((5x+2)^6\right)$

r $\log\left((x+6)^5\right)$

s $\log_2(5x)$

t $\log_{10}\left(\dfrac{2}{x}\right)$

u $\log\left(m^2\right)$

v $\log\left(5x^{\frac{2}{3}}\right)$

w $\log\left((14x)^{\frac{1}{3}}\right)$

x $\log\left(\sqrt{\dfrac{c^8}{d}}\right)$

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4 Rewrite the following as the sum or differen[illegible] logarithms without any powers, fractions or radicals:

a $\ln\left(\frac{15}{8}\right)$ b $\ln\left(\frac{pq}{r}\right)$ c $\ln\left(\frac{19}{7}\right)$

5 Rewrite each of the following as a single logarithm or integer:

a $5\log x^3 - 4\log x^2$

b $5\log x + 3\log y$

c $8\log x - \frac{1}{3}\log y$

d $7\log x - \log\left(\frac{1}{x}\right) - \log y$

e $7\log_{10} 5 - 21\log_{10} 25$

f $5\log_{10} 8 - 3\log_{10} 4$

g $2\log_6 3 + \frac{1}{3}\log_6 64$

h $\log_2 36 - 2\log_2 3$

6 Rewrite each of the following expressions as a single logarithm:

a $\ln 3 + \ln 5$

b $\ln 24 - \ln 4$

c $\ln 5 - \ln 7$

d $\ln 11 + \ln 5 + \ln 8$

e $\ln 24 - (\ln 2 + \ln 6)$

f $\ln 7 + \ln 2 + \ln 3 - \ln 5$

7 Rewrite each of the following expressions as a single logarithm:

a $6\ln x - \frac{1}{6}\ln y$

b $\ln(3x) + \ln(5y)$

c $\ln\left(x^2\right) + \ln\left(x^3\right)$

d $3\ln\left(x^5\right) - 4\ln\left(x^2\right)$

8 Use the properties of logarithms to express each of the following without any powers or radicals:

a $\ln y^2$ b $\ln\frac{1}{y^2}$ c $\ln\sqrt{y}$ d $\ln\sqrt[3]{y}$

e $\ln y^{-5}$ f $\ln y^{\frac{2}{9}}$ g $\ln y^{\frac{1}{2}}$ h $\ln\sqrt{w}$

9 Use the properties of logarithms to rewrite each of the following in the form $\ln b^a$ where $a > 0$:

a $-2\ln x$

b $\frac{\ln x}{2}$

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Change of base

10 Rewrite the following in terms of base 10 logarithms:

a $\log_4 16$

b $\log_3 0.9$

c $\log_3 \sqrt{5}$

d $\log_a B$

11 Rewrite $\log_3 20$ in terms of base 4 logarithms.

12 Consider the following logarithmic expressions:

i Rewrite the expression in terms of base 10 logarithms.

ii Now, evaluate each to two decimal places.

a $\log_8 21$

b $\log_4 \sqrt{5}$

c $\log_2 3^3$

d $\log_\pi 105$

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